

# Crime Analysis

## Progress Report

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### Abstract

A concern amongst every citizen in the world is their own safety. One of the major factors that affect citizens' perceived safety is the level of crime around them and how it fluctuates along the years. Therefore, this research aims to analyse crimes using different data mining techniques to aid law enforcements mitigate crime. New York City's crime data from 2006-2019 was used for this research. By applying supervised machine learning algorithms, we intend to observe how future crime rate will vary and to predict the law category type of a crime with its features. Moreover, association rule mining is performed to identify frequent occurring patterns and relationships in our dataset along with clustering techniques to profile criminals to crimes and vice-versa. This research paper will also serve as a comparative study to other published studies.

**Keywords:** Crime Analysis, Data Mining, Crime Prediction, Crime Profiling, Crime Patterns, Machine Learning, Artificial Intelligence, NYC Dataset

## 1 Introduction

Crimes in urban spaces are a concern for every person's own welfare. Law enforcement agencies continuously strive to understand the nature of crimes which are occurring in their responsible city with the aim of: identifying who is committing them; where and when they are most likely to occur; what is the likelihood that a certain crime takes place; and how much will the rate vary along the years. This process is vital in any agency to protect the safety of their citizens [1].

In our research, we attempt to tackle this issue by using different data mining techniques related to the field of Artificial Intelligence (AI) to help these agencies understand their data well and automatically. This would help them develop a preventive plan to reduce the amount of crimes.

### 1.1 Motivation

The topic of crime is a topic which of global interest since it concerns each and every one of us, particularly in places which are densely populated. A recent article published by BBC <sup>1</sup> highlighted the fact that USA has been seeing an increase in crimes, especially 'Violent' crimes with New York being at the forefront of this report. Subsequently, it was

thought and decided that analysing such a highly popular and populated city, like New York City (NYC) within such a state that is seeing a significant number of crime on the increase yearly would be a great domain for this research purpose. Moreover, the availability of comprehensive and detailed datasets on crimes in NYC also renders such analysis more promising.

### 1.2 Why the Problem is Non-Trivial

This research has a number of challenges to be addressed. First of all, dealing with large datasets can be challenging at times. This is because creating machine learning models can be computationally expensive when having an enormous amount of data involved to run on a normal computer machine, like ours.

Apart from this, finding the right dataset for our particular use case is an element of concern. Different countries collect different data since crime information is considered sensitive, and this collection is different in every country. Additionally, not all countries publish their crime data and thus makes it more difficult for us to find the most suitable data. Besides this, datasets are sometimes fragmented and would need to be aggregated together. This aggregation can be quite challenging because unique keys from the fragmented databases are either omitted or contains empty values, as happened with the London dataset which will be explained later on. It is also vital to ensure that any available datasets are not only comprehensive but are also ideal for our research topic objectives.

Another issue would be that any unsupervised learning technique used is difficult to evaluate it. The reason behind this is that the models generated cannot be evaluated directly due to lack of ground-truth data. Moreover, from the background research conducted, none of the authors actually explain how they went about with their evaluation strategy when using such algorithms.

The rest of this paper is organised in the following manner. Section 2 outlines our aims and objectives of this research whilst Section 3 describes in depth the research carried out on what other authors have done in their research. Subsequently, we describe how the solutions of other researchers have influenced the scope of this research paper. Section 4 then delves more into what we intend to achieve in this paper, and how it will be achieved. Therefore, we describe the dataset chosen and the objectives of this paper. Subsequently, Section

<sup>1</sup><https://www.bbc.com/news/57581270>; last accessed 3rd December 2021

5 highlights the evaluation strategy which we will carry out for our models created whilst Section 6 concludes this research in summary.

## 2 Aims and Objectives

The main aim for this research is to use different data mining techniques on an urbanised city crime dataset to profile and find patterns between suspects, victims, crimes and other features, predicting the law category likelihood as well as predicting the future crime rate.

We decided to set the following objectives to reach our research aim:

- Generating statistical insights and explanatory analysis to gain insights into the chosen dataset.
- Crime profiling to identify associations between crime data, suspects and victims using clustering techniques and association rule mining.
- Predicting the category of law enforcement of a crime given crime details.
- Predicting if the number of crimes will continue to grow or cease with time using time series forecasting methods.

## 3 Background Research and Literature Review

From the background research conducted from other researchers work, the application of different data mining techniques is vast for crime analysis and can take many different forms. Below is a summary of the overall objectives which authors have performed in their research study:

- **Crime Hotspot Detection and Prediction:** finding locations having a high degree of crimes and predicting crime. Prediction can either be a classification or regression task. [3, 6, 8, 10, 12, 15, 16, 20]
- **Crime Patterns and Profiling:** finding hidden information and similarities between crimes, suspects and victims [2, 5, 7, 12, 16]
- **Criminal Network Analysis:** finding common relationships/patterns/ties between criminals to link them together. [20]
- **Statistical and Exploratory Analysis:** generating statistical insights to provide an overview of what the dataset contains. [15]
- **Police Patrolling:** Assigning and allocating police forces to specific locations and times. [20]
- **Near Repeat Victimisation:** identifying the risk of a person being victimised at a given place and time [20]
- **Next-Location Prediction:** Predicts the location where the offender is likely to commit another crime based on their historical crime data. [20]

Zhao and Tang research paper [20] focuses on giving an explanation on the different types of objectives which can be

implemented for crime data rather than implementing them themselves.

The objectives chosen for our research paper are crime predictions, crime profiling and patterns, crime hotspot detection as well as statistical insight analysis. These objectives will be discussed in more detail below by highlighting what other researchers have carried out in their work.

### 3.1 Predicting Crimes and Hotspots

Crime prediction is perhaps one of the most popular objectives which researchers aim to attain within this domain [3, 6, 8, 10, 12, 16–18]. This objective is often handled by using supervised machine learning algorithms, and can come into two forms, either as a classification task or as a regression task. A classification task is one which uses a model that classifies a discrete value whilst a regression task is one that utilises a model to predict some continuous value. Hence, this objective is divided into two separate sub-sections as seen below.

**3.1.1 Crime Type Prediction.** This classification problem is usually solved by using various supervised machine learning algorithms.

In the works of [3, 6, 8, 10, 15, 16], it is interesting to note how these authors uses different algorithms to predict the type of crime or location.

For example, the authors in [15] uses Naïve Bayesian classifier and decision tree classifier as algorithms. They used an open-source dataset consisting of crimes in each of San Francisco's districts<sup>2</sup>, having temporal and spatial features together with the crimes. Their purpose is to predict the type of crime in a particular location at specific times. Similar to the works of [15], Sardana et al. [16] implemented similar solutions. The main differences between these two papers is that the latter groups the crime types according to the corresponding law category of USA and aims to predict this categorical variable. The reason why they do this is because higher granularity in the crime type can be much more challenging to predict, thus predicting its law category of the crime leads to a more accurate model. Moreover, these authors use more machine learning algorithms such as Logistic Regression, Random Forest and Gradient Boosting. Therefore, a comparative study was done in search of trying to find the best accurate model for their problem.

On the other hand, [6] implemented similar solutions as well, but decided to integrate additional data to crime. Apart from crime, temporal and spatial data, Elluri et al. also decided to include weather data. They focused on NYC crime data [13]. This dataset includes all data like the aforementioned SF dataset, but also includes data related to the victim and suspect (age group, race, gender respectively). Their

<sup>2</sup><https://www.kaggle.com/roshansharma/sanfrancisco-crime-analysis/data>; last accessed 3rd December 2021

main aim is to study if weather data can help in predicting the type of crime. Hence, they did a comparative study between predicting crime types with and without weather data. Additionally, they went deeper in their implementations by implementing more machine learning algorithms like Support Vector Machines (SVMs) and deep learning methods, including NNs, CNNs and RNNs.

In the paper of [17], Sathyadevan et al. collected an unstructured dataset from news, websites, social media and other means. Then, they used the Naïve Bayesian classifier to predict the type of crime given the unstructured document. This is very interesting to note since they managed to build a 90% accurate model that is able to classify crime type from such data. Apart from this, they also implement a predictive model which is able to classify whether a crime will likely occur in a given place according to some attributes. In this task, the Decision Tree Classifier is used since it is very easy to interpret with a high level of accuracy.

**3.1.2 Crime Rate Prediction.** This prediction is another common implementation done for crime analysis.

In the works of [12], Marzan et al.'s main aim is to predict the future crime trends. They focus on predicting the number of crimes in each district of Manila City's crime data for the future days and weeks. This is achieved by using Time Series Forecasting methods, a data mining technique which is dependent of time. These authors implement several machine learning and deep learning algorithms that predict future crime rates which include Linear Regression, Gaussian Processes, Multilayer Perceptron and SMOreg. They evaluate their models with each other using Mean Absolute Percentage Error as a forecasting accuracy measure.

On the other hand, Wang et al. [18] moved on from the traditional use of demographic and geographic data to predict future crime rates. Therefore, they make use of Point-of-Interest (POI) and taxi flow data in addition to the aforementioned traditional data to predict the expected crime rate more accurately. POI data is data that provide venue information such as coordinates, category, popularity and reviews. Recent studies have highlighted that using such POIs data are useful to profile neighbourhood functions [19] which could have a correlation with the crime rate. Taxi flow data reflect how people commute in the city and subsequently, can highlight frequently locations travelled to which in turn can be tied to the location being examined. The main reason why including these type of data is to improve the overall accuracy of the developed models.

All papers discussed in this section use similar evaluation strategies. For classification problems, the classification report containing F-1 score, recall, precision and accuracy for each model is outputted. Sometimes, this evaluation strategy is used in conjunction with the Log Loss function to measure the performance of the model. On the other hand, for regression problems, the most common evaluation strategy

is the Mean Absolute Percentage Error (MAPE), however, other metrics can also be computed, for example, the Mean Squared Error (MSE).

### 3.2 Crime Pattern Discovery and Profiling

Another objective which is commonly implemented within most crime analysis research papers is crime pattern discovery and profiling. This topic is usually handled through clustering techniques or association rule mining techniques. The former technique is more used for profiling crimes, victims or criminals whilst the latter technique is mainly used for deducing frequent occurring crime patterns.

Gera and Vohra [7] uses k-Means clustering algorithms to create crime profiles for the city of Delhi, India by analysing the results from the clusters generated. On the other hand, Adderley and Musgrove [2] aimed to identify the typical profile of a sexual offender by applying a Self-Organising Map (SOM) that performs clustering. Similarly, the authors in [9] also makes use of SOMs to cluster their data. However, their main objective is to profile crimes with criminals to unsolved crime incidents from the clustering results. A Multi-Layer Perceptron (MLP) neural network is included on top of the clustering as they believe that it will benefit in crime matching, i.e. classifying crime to criminals and vice-versa. The main reason why the above mentioned authors are using SOMs is because this implementation is useful in reducing high-dimensional data to a lower dimensional [11][9]. Moreover, it creates a 2D topological map which can show links between neighbouring clusters.

In the works of Boo and Alahakoon [5], they used Growing Self-Organizing Map (GSOM) - a hierarchical clustering technique, an extension to a SOM. The main aim of these authors is the explore hidden patterns at different granularity of the data. Hence, a GSOM fits perfectly to their implementation as it allows them to highlight clusters with its respective details with higher granularity. Additionally, GSOMs are a great way to visualise your data and understand it well. In [5], an example is shown whereby they are showing the different types of firearms which they found in the clusters generated.

Crime patterns can also be achieved using association rule mining methods. These unsupervised learning algorithms are basically used to find frequently occurring relationships between hidden features in the dataset. In the paper of Marzan et al. [12], the Apriori algorithm was used as an association rule mining technique to find patterns between crime type, occurrence time, weather, holiday, the crime location and the district. This algorithm generates a set of rules according to some predefined minimum support and minimum confidence having the district variable as the consequent. The authors only output those generated rules having their lift value greater than 1, meaning there is a positive correlation. Hence, using this technique, Marzan et al.

found hidden relationships which can be used by law enforcement to prevent crime according to the above rules. The same technique is used in [17]. Here, the Apriori algorithm is used to find the patterns and trends of crimes happening in specific locations. Hence, using these rules generated for each location, it can help authorities in deducing that there is a probability that a crime might occur if it follows the rules for that locality.

## 4 Proposed Solution

As described before, our proposed solution will mainly be focusing on crime prediction (both classification and regression) together with crime profiling. These will be each explained in the below sub-sections by highlighting the dataset considered in our research, the approach taken and the papers deemed influential in our objective's decisions.

### 4.1 Datasets

This research paper aims to create insights and making sense of the data through data mining techniques. Hence, choosing the right dataset containing various different attributes to achieve this is challenging. From the background research conducted, some authors [7, 12] collected their data manually for their own need only. This made it more difficult to find such diverse datasets. Moreover, every country has its own way on how it collects, stores and publishes its data to the general public for analysis.

Initially, it was considered to use the Police API of UK<sup>3</sup> to collect crime data. However, it was then discarded since this dataset had multiple shortcomings in the amount of features it contained. This dataset was partitioned into two segments, i.e. street crime data and stop & search data. The former only contained data about the year and month the crime occurred, the type of street crime (ex: theft) and its outcome whilst the latter contained date (high granularity including time), object searched, officer as well suspect's details (ex: age and ethnicity) and outcome. However, these two datasets could in no way be integrated together, hence, we couldn't use all its features to solve our problem. Moreover, other drawbacks included that only data from 2018 to date could be retrieved from the API and that it was very challenging to integrate additional data, like weather data due to merging difficulties.

Therefore, the dataset which will be used in this research paper is about all of NYC's complaints [13], the same dataset utilised by [6]. Every row in this database is a complaint, which is ultimately a reported crime. This dataset contains numerous distinct features including *date*, *time*, *offense description*, *offense law category*, *borough*, *victim details*, *suspect details*, *outcome* and more. Moreover, additional data was integrated to [13] including weather data using OpenWeather's

API [14] as well as other data such as median income, population and enrollment<sup>4</sup>.

The dataset was then pre-processed in the following way:

1. **Data Reduction:** removing unneeded columns as well as those columns having a significant percentage of *null* values.
2. **Data Cleaning:** dealing with missing and inconsistent values. Fixing column's values which are misspelled or not easy to understand.
3. **Data Integration:** integrating new attributes from current features, similar to [12]. For example, the date field is split into *hour*, *day*, *month* and *year*. Additionally, *isHoliday* feature was integrated according to a predefined list of all the holidays of NYC.
4. **Data Transformation and Discretisation:** the data integration part ended up with features having multiple distinct values, which are not ideal to solve our problem with high accuracy. Hence, temporal data was transformed by mapping their value to their respective group. Therefore, this process created additional columns including *hour group*, *day group* and *season*. Moreover, all categorical features were encoded prior to creating the models for prediction.

### 4.2 Crime Type Prediction

We intend to perform crime type prediction utilising similar methodologies utilised by Elluri et al. [6], Pradhan et al. [15], and Sardana et al. [16]. This problem will be tackled as a multi-class classification problem using various supervised learning algorithms. The motivation behind this objective is to develop a highly accurate model which can classify the law category given temporal, spatial, weather and related crime data. It is important to note that the same offences can yield to different law category outcomes, for instance, the *Dangerous Drugs* crime type can either have *Misdemeanour* or *Felony* as its law category, depending on the grievousness of that crime. Therefore, this objective will aid law enforcement to predict the law category based on its features.

Currently, we have trained multiple machine learning algorithms that are predicting this target variable such as *Decision Trees*, *Random Forest Classifier*, *kNN* and more. All models are currently being fed the same features, including *weather data*, *temporal and spacial data* and *relevant crime data* (*suspects*, *victims*, *offense type*, etc). Weather data was augmented with the main scope and aim to understand if weather data helps in predicting the law category of a crime. Implementing multiple models is done in order to compare the accuracy of each model created and choose the best model that fits our data. Up to now, accuracy is ranging from 68% to 81% and these results will be compared to other research studies that implemented similar systems [16]. Therefore, our next

<sup>3</sup><https://data.police.uk/data/>; last accessed 3rd December 2021

<sup>4</sup><https://data.cccnewyork.org/data/download#0/0>; last accessed 3rd December 2021

objective would be to try and improve this accuracy score by determining those features which are highly influential to the target variable rather than just pitching all attributes.

### 4.3 Crime Rate Prediction

This objective will be solved using time series forecasting algorithms, similar to what Marzan et al.'s [12] have done. Regression-based machine learning models in combination of different features will be used to predict the expected future crime rate per NYC's borough yearly.

Currently, we are predicting next year's crime rate based on previous yearly crime rate, irrespective of the borough. Examples of models built are *Linear Regression*, *Decision Tree Regression* and *Random Forest Regression*. Moreover, the Mean Absolute Percentage Error (MAPE) is being used as an error metric to determine the most accurate model and from the implementations so far, this error is ranging from 18% down to 6.5%.

Similarly to crime type prediction, we want to perform a comparative study to try and further improve this error. Hence, all models will be re-built with different features and analyse how this is contributing to the accuracy of each model. Therefore, we intend to use the *Population*, *Median Income* and *Enrollment* features as part of our models and compare them.

### 4.4 Crime Profiling

The task of crime profiling was used to find background information and hidden patterns about the crime dataset. Following the implementation of Gera and Vohra in [7], this task will be handled using k-Means clustering algorithm. So far, we have clustered the whole dataset using all its features excluding temporal data. Then, heat-maps are generated by plotting clusters in the x-axis and the feature's values on the y-axis, as done by [10]. These heat-maps will highlight interesting associations between data within clusters. However, from the work implemented up till now, no interesting facts can be deduced from these visualisations since crimes within this NYC dataset seem to be taking place randomly, irrespective of its features. Given the current situation, we still intend to pursue this objective with a more thorough investigation.

## 5 Evaluation Plan

From the literature review conducted, when it comes to clustering and association rule mining, most of the authors did not explain how they evaluated their results. The only who did so was Adderley and Musgrove in [2] whereby they communicated with a highly skilled crime analyst. However, although this would be great to carry out in this research, finding an analyst to be involved in such a research is difficult.

Seeing this, it was decided to follow similar evaluation methodologies as done by Barbara [4]. In her dissertation, Barbara evaluated their Victim Profiles by dividing the dataset into Training set and Test set. Then, the aforementioned algorithms are implemented on the training set and explanatory analysis is carried out. The same methodology was applied on the test set, and if the same analysis and trends are deduced from the test set, then it means that the profiles generated and conclusions inferred are not overly specific to the data subset used. Essentially, this means that in our case, clusters will be generated for the years of 2006-2018, and since we have available data for 2019, then the actual clusters for 2019 will be compared with other clusters generated.

Another approach which can be taken is that multiple experiments having different features can take place. Clustering will be done in each experiment and if clusters do not change with and without a particular feature, then that feature is considered to have no effect on the profiles.

Otherwise, if the above will not be done, then all models are run on the whole dataset and explanatory analysis will be performed as a whole.

The evaluation strategy for the supervised machine learning algorithms is straightforward since predicted values can be compared to the actual labelled values using several metrics. This can be achieved by outputting the *Accuracy*, *F1-score*, *Recall* and *Precision* for each created model, in the case for classification models as done in [6, 10, 15, 16]. On the other hand, for the regression models, different metrics can be computed to determine the error between the actual value and predicted value. These include *Mean Absolute Error (MAE)*, *Mean Absolute Percentage Error (MAPE)*, *Mean Squared Error (MSE)* and *Root Mean Squared Error (RMSE)*. In our case, the *MAPE* forecasting measure accuracy is used, the same metric utilised by Marzan et al. in [12]

## 6 Conclusion

This research summarises what has been implemented so far in pursuing the described research aims and objectives and all challenges faced so far. The goal of this paper is to perform crime analysis on an urban city using AI driven techniques, as explained in section 4 above.

Up till now, we came across several challenges, mainly being finding the right dataset for our research purpose as well as searching and aggregating other data that can be useful for our AI models. Another challenge which is still being tackled is that of Crime Profiling where we intend to survey this objective deeper.

On the other hand, we have managed to implement the Crime Prediction objective by already having several supervised machine learning models created. This goal will be further implemented with the aim of using additional data to try and improve the model's accuracy with regards to crime rate prediction whilst in crime type prediction, we

aim to identify and use only those features which highly correlate with the target output to also improve accuracy. Subsequently, in both components, a comparative study can still be performed. Finally, all models created will be evaluated accordingly to deduce the efficacy of our research.

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